

OEL (Self-Learning & Implementation)

Topic: Graph Data Structure in C++

Objective

The objective of this OEL is to enable students to independently learn, implement, and analyze Graph data structures using C++. Students will explore theoretical concepts through self-study and apply them practically through coding.

Self-Learning Tasks

Students are required to self-study the following concepts: definition of graph, basic graph terminology, types of graphs (directed, undirected, weighted), graph representations (adjacency matrix and adjacency list), and graph traversal algorithms including BFS and DFS.

Implementation Tasks (C++)

Students must individually implement the following programs in C++: graph representation using adjacency matrix, graph representation using adjacency list, BFS traversal using queue, and DFS traversal using recursion or stack. All programs must be properly commented.

Submission Instructions

Students must write all programs independently and submit their complete work via email.

Email: asma.javaaid@uajk.edu.pk

Submission Deadline: 16-01-2026 (on or before 04:00 PM)

Expected Learning Outcomes

After completing this OEL, students will be able to understand graph concepts through self-learning, implement graph-based solutions in C++, apply BFS and DFS algorithms, and demonstrate improved problem-solving skills.